

EARTH 211 – Spring 2026

Final Report – due June 12th, 2026

Investigating the potential of beneficial microbes—Agrobacterium and Arbuscular Mycorrhiza—on mitigating the effects of Drought and high Temperatures, on Black Seed (and Cabbage) Plants

Introduction — As climate change rapidly worsens, intensified drought and heat make it harder for plants to grow in many parts of the world, especially in desert regions like Saudi Arabia's Empty Quarter (Al Rub'a Al Khali) [1]. This research project explores the impact of beneficial microbes—Agrobacterium (a bacterium) and Arbuscular Mycorrhiza (a fungus) [2,3]—on the growth of *Nigella sativa* (commonly known as Black Seed or Habbat al Barakah) under these harsh conditions. *N. sativa* is culturally, medicinally, and culinarily important in Saudi Arabia. However it typically grows in cooler areas with more rain, and struggles to survive due to hotter, drier environments [4]. The two research questions I am addressing in this project are **(1) How do beneficial microbes affect the height and qualitative survival indicator of plants under climate stress? & (2) What is the impact of microbial inoculation on chloroplast levels and leaf counts in stressed plants?** The significance here is that—if successful—this approach could offer a sustainable and scalable way to grow crops in desert environments, helping conserve plants for future generations. The findings could also benefit other arid regions and significant plants facing similar climate challenges [2,3].

Data — For this project, I used “*Nigella sativa* & *Brassica oleracea* resilience towards environmental stressors while subjected to microbial Inoculation.” This is data I collected during my independent research project at the Chicago Botanic Garden (Summer 2025), under the mentorship of Prof. Alicia Foxx. The plants used in this study were grown from seed in a controlled Growth chamber at the CBG, where the plants underwent a 5 week growth period following 2 weeks of germination. The specific seeds utilized were collected from the garden itself, Jordan, and Idaho USA. Originally this project was supposed to be exclusively experimenting on *Nigella sativa*, however, 2-weeks into the experiment, the plants’ germination rate was low and *Brassica oleracea* seeds were planted to make up for the anticipated loss of individuals to carry out the experiment on.

These are the variables used throughout this project:

- a. rack: a physical location and an indicator of treatment type,
- b. species: Factor tracking plant varieties (*N. sativa* and *B. oleracea*).
- c. treatment: Factor tracking 12 distinct environmental/microbial stress cohorts.
- d. height & final_height: Continuous quantitative shoot height (cm), last measured on 08/29.
- e. status & final_status: Categorical qualitative survival score (0=Dead, 1=Necrosis/Wilting, 2=Thriving).
- f. chlorophyll & chlorophyll_final: Continuous quantitative photosynthetic measurement.
- g. leaf_count & leaf_count_final: Discrete count of emerged true leaves.

To get this dataset from its raw state to a cleaner version that is much suitable for data analysis, some cleanup and processing was needed. First of all, there were several entries with mixed data types: a string where only numbers were expected. Next, due to the addition of *Brassica oleracea* to the dataset and the fact that this project was not focused on species-specific changes since it is more preliminary, both species were grouped under the same label for analytical purposes. Additionally, due to a high mortality rate—a natural outcome for a plant-stress experiment—a large number of data cells were left unpopulated. Any

kind of “non-entry” cell (meaning not containing its intended value(s)) was unified under the assignment of “NA.” Moreover, the fraction of living plants were placed in a subset of their own, to avoid any data skewing due to dead individuals. Finally, the clean data subsets were stored in the “03_ProcessedData (Recent)” file.

Methods

1. 95% Confidence Intervals (CI) — This method was used to estimate the true mean final height of surviving plants. It also served as an overall indicator of the plants’ response to environmental stressors. Variables used here were standard to the way a CI is calculated.
2. One-Way ANOVA — This method was used to analyze the extent to which mean plant height varied based on which treatment group it was subjected to. Here, a one-way ANOVA was appropriate since a continuous dependent numeric value (plant height) was being compared to independent categorical variables (treatment group). Here, the null hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3 \dots = \mu_n$ (where n is the number of treatments) and the alternative hypothesis H_1 : is that at least one of the mean plant heights will be different from the rest.
3. Tukey’s HSD Post-Hoc Test — This method came naturally following ANOVA. It was used to find which treatment pair(s) have statistically significant height difference.
4. Chi-Square Test of Independence (with Monte Carlo Simulation) — This method was used to evaluate the relationship between treatment group and final plant status. Monte Carlo simulation is specifically needed here because Chi-Square on its own in this case is not successful—due to the relatively small size of viable data points available—therefore, a Monte Carlo simulation with 2,000 replicates solves this problem and provides us with a more-accurate p-value [5]. Here, the null hypothesis H_0 : plant health status is independent of the treatment group and the alternative hypothesis H_1 : is that plant health status is, in fact, dependent on the treatment group.
5. Spearman Rank Correlation — This method was used to measure the relationship between final plant height and final chlorophyll level, in order to determine how closely related (and in what direction) are these variables. A non-parametric correlation was performed here as opposed to a parametric one, due to the non-linearity of the data, due to its biological nature. Here, the null hypothesis H_0 : final plant height and chlorophyll levels are not correlated whatsoever ($\rho = 0$), and the alternative hypothesis H_1 : is that there is significant correlation between final plant height and chlorophyll levels ($\rho \neq 0$).

Results — Present your main findings using text, tables, figures, maps, or model outputs. Do not only paste output from R or Python. Explain what the results mean.

1. 95% Confidence Intervals (CI): (7.573742, 8.600803)
This confidence interval means that if a random plant sample (the same size and from the same dataset) was chosen several times, there is a 95% chance that the true population mean of plant heights is included in the above interval.

Table 1. This table contains the variables and their values, as used for 95% CI calculation.	
Metric	Value
Mean	8.087273
SD	1.943084
N	55
Lower_CI	7.573742
Upper_CI	8.600803

2. One-Way ANOVA: The p-value = 0.00089 here is much smaller than 0.05, meaning that there is enough evidence to reject the null hypothesis.

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Df Sum Sq Mean Sq F value Pr(>F)
treatment  11  99.54   9.049   3.729 0.00089 ***
Residuals  43 104.34   2.427
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

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3. Tukey's HSD Post-Hoc Test: Results from this method show 4 interactions possessing statistical significance $p < 0.05$:
 - a. Control (Fungi) vs. Control (Bacteria & Fungi) $p = 0.0203$, Plants given only Fungi were significantly shorter than plants given both Bacteria and Fungi.
 - b. Drought vs. Control (Fungi) $p = 0.017$, Drought-stressed plants were significantly taller than control plants only given Fungi.
 - c. Drought + Bacterial Inoculation vs. Control (Fungi) $p = 0.0123$, Drought-stressed plants inoculated with Bacteria grew significantly taller than control plants only given Fungi.
 - d. Drought + Combined Microbial Inoculation vs. Control (Fungi) $p = 0.0060$
Drought-stressed plants inoculated with both Fungi and Bacteria were significantly higher than control plants only given Fungi.
4. Chi-Square Test of Independence: X-squared = 3.5714, df = NA, p-value = 0.1229. Because $p > 0.05$, the result of this test is statistically insignificant, and we fail to reject the null hypothesis. This means that we cannot with proper certainty, correlate qualitative plant health status, with the treatment groups used throughout this project.
5. Spearman Rank Correlation: $S = 160.99$, p-value = 0.9468, rho = 0.02431622. Because $p > 0.05$, even though rho is not 0, the result of this test is statistically insignificant, and we fail

to reject the null hypothesis. This indicates that there is virtually no linear correlation between final plant height and final chlorophyll levels among the surviving cohorts.

6. Final Height across Treatments Boxplot

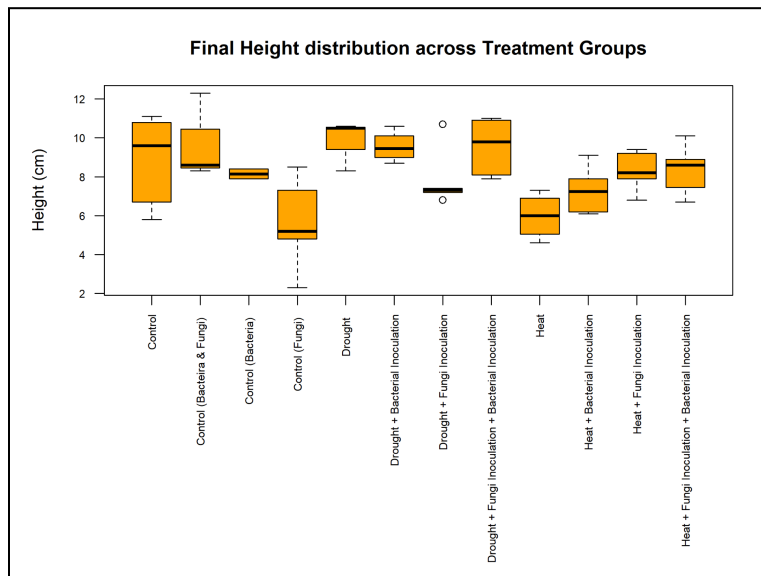


Figure 1. This figure represents several side-to-side boxplots of plant final height with respect to the various treatment groups used throughout this project. Looking at the Controls, Control (Fungi) resides generally lower than the other groups, alluding to the fact that under normal environmental conditions, only using fungal inoculants can stunt plants' growth instead of supporting it. Looking at drought treatment variations, the control, bacterial inoculants, and both fungal and bacterial inoculants seem to have similar effects—whereas once more, the fungal group underperforms. Finally, when it comes to heat treatment variations, we actually see the opposite. Fungal inoculants show best performance, followed by the fungal+bacterial combination, bacterial inoculation, ending with the control group.

7. Height vs Chlorophyll Levels Scatter Plot

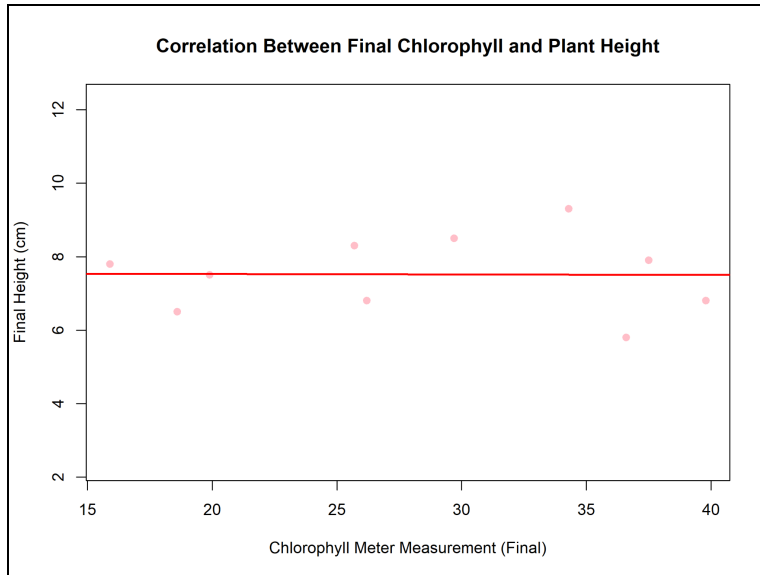


Figure 2. This figure is that of a scatter plot representing the relationship between final chlorophyll and plant height. The almost entirely horizontal trendline shows that there is practically no correlation between these two variables.

Discussion —The goal of this experiment was to better understand the relationship between plant resilience against climate stressors (heat & drought here) and microbial inoculation of bacteria and fungi, specifically in terms of height, health and chlorophyll count. Initially, the One-Way ANOVA output ($p=0.00089$) indicates that there is a statistically significant correlation between plant heights and at least one of the treatment groups. To further explore this, the post-hoc Tukey HSD test shows that in control and drought conditions, there is statistically significant evidence pointing to the fact that fungal inoculation is stunting plant growth in the aforementioned cases, instead of supporting it. This aligns well with the results seen from Figure 1, the “Final Height across Treatments Boxplot.” This plot implies that isolated fungal inoculants in normal (unstressed) environmental conditions, might be competing for resources or interfering in a different way with the plants’ growth instead of forming a symbiotic relationship with them as expected. On the other hand, under stressful environmental conditions of drought, both sole bacterial inoculation and combined bacterial and microbial inoculation are shown to improve plant resilience. Moreover, since the Chi-Square test output was statistically insignificant, we fail to reject the null hypothesis, and therefore aren’t able to correlate the treatment group with qualitative plant health. While real-life first-hand observations did indicate that a correlation between those two variables might be present, the low number of samples at hand makes it not possible to declare that those observations were not random. Another statistically insignificant result was that of Spearman Rank Correlation, meaning that we are also unable to—with statistical confidence—correlate plant height and chlorophyll levels. This result is also supported by Figure 2, in which we see a near entirely horizontal slope, showing that there is virtually no correlation between a plant’s height and its leaves’ chlorophyll content. The 2 main limitations (can be viewed as one) associated with this project in my opinion are the low germination rate of seeds early on in the experiment, followed by a high mortality rate later on. This resulted in a loss of sample size, which affected the kind of

analyses that I was able to perform, as well as their statistical significance. This limitation also led to the introduction of a plant species outside of the scope of the original research proposal. This led to an additional possible revenue of error: undetected difference in experimental values as a result of species differences. The high mortality rate also led to survivor-bias, since the represented data is only from groups that were already resilient enough to survive the several week environmental stressor experiment. If given more time and/or resources, this project can be improved by increasing the sample size (through planting more seeds and providing them with better germination conditions), extending the timeline of the project to observe long term changes, and potentially use higher-end technology to deliver more accurate environmental condition(s) changes to the plants.

Conclusion — To summarize, this study successfully contributed to better understanding the relationship between plant resilience against climate stressors (heat & drought here) and microbial inoculation of bacteria and fungi. Despite some of the methods used providing statistically insignificant results, some indicate that to an extent—in need of further research—microbial inoculations *do* mitigate the effects of climate change (represented by heat and drought) against the plants in question.

References (IEEE Format)

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- [4] M. A. Foxx, A. M. Iyer-Pascuzzi, and M. L. Bowles, "Cultivars and wild plants differ in trait variation with surprising root architectural trait conservation," *Restoration Ecology*, vol. 30, no. 6, p. e13336, 2022. [Online]. Available: <https://doi.org/10.1111/rec.13336>. [Accessed: Mar. 14, 2025].
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Generative AI Acknowledgment

I briefly used Google AI throughout this project. The meaningful ways in which it was employed are the following:

1. General debugging (specifically in handling the mixed type data inputs and successfully running the Chi-Square Test of Independence with Monte Carlo Simulation).
In this regard it was mostly helpful, though was only used after a general google search and a browsing through forums.
2. Assistance in executing the spearsman test. It was helpful here, since I wasn't sure how to structure the code for this one.
3. Initially brainstorming appropriate data analysis methods, considering the state of my data. Every suggested method was searched about, and then the top ones deemed relevant and useful by me were actually used.

****Note:** all AI outputs were double checked by me and proper external resources were cited.